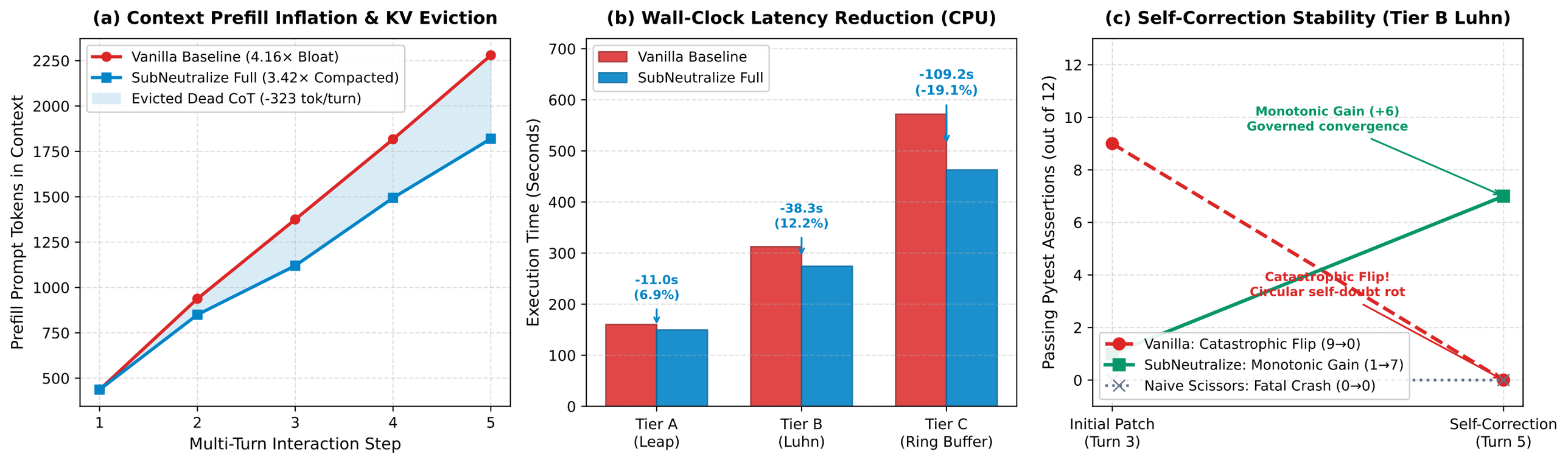
(a) Context Prefill Inflation & KV Eviction
Vanilla Baseline (4.16× Bloat)
SubNeutralize Full (3.42× Compacted)
Evicted Dead CoT (-323 tok/turn)
Prefill Prompt Tokens in Context
Multi-Turn Interaction Step
(b) Wall-Clock Latency Reduction (CPU)
Vanilla Baseline
SubNeutralize Full
-11.0s
(6.9%)
-38.3s
(12.2%)
-109.2s
(-19.1%)
Execution Time (Seconds)
Tier A
(Leap)
Tier B
(Luhn)
Tier C
(Ring Buffer)
(c) Self-Correction Stability (Tier B Luhn)
Monotonic Gain (+6)
Governed convergence
Catastrophic Flip!
Circular self-doubt rot
Vanilla: Catastrophic Flip (9→0)
SubNeutralize: Monotonic Gain (1→7)
Naive Scissors: Fatal Crash (0→0)
Passing Pytest Assertions (out of 12)
Initial Patch
(Turn 3)
Self-Correction
(Turn 5)